

ASSIGNMENT - 1

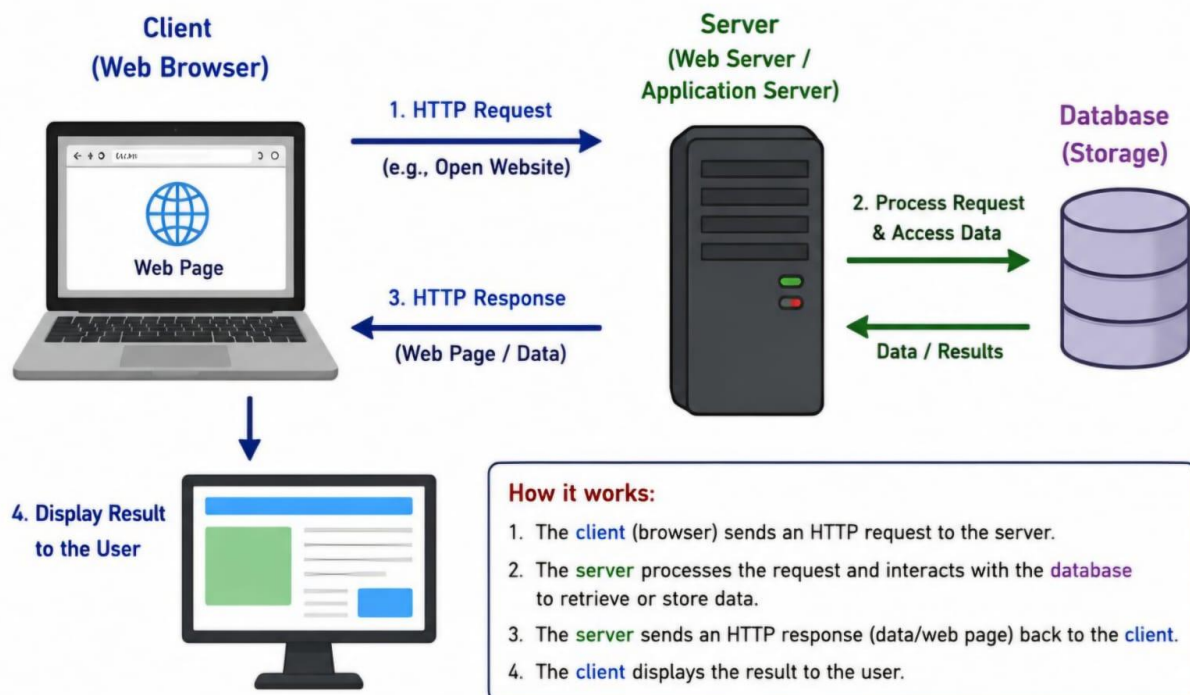
Ques 1 . Explain the difference between frontend, backend, and full-stack development with suitable real-world examples.

Ans 1 .

Frontend	Backend	Full-Stack
Develops the user interface (UI).	Develops server, database, and business logic.	Develops both frontend and backend.
Users can see and interact with it.	Works behind the scenes.	Handles the complete web application.
Example: Login page, buttons, website design.	Example: Login authentication, database, payment processing.	Example: Builds the entire Amazon/Flipkart website.

Ques 2. Create a simple diagram showing how the client-server model works in web architecture.

Client-Server Model in Web Architecture



Ques 3 . Describe how a browser requests and displays a web page from a web server.

Ans 3.

- User enters the website URL in the browser.
- The browser sends an **HTTP/HTTPS request** to the web server.
- The server processes the request and retrieves data (if needed).
- The server sends an **HTTP response** back to the browser.
- The browser renders and displays the web page.

Ques 4. Identify and list the tools required to set up a web development environment. Explain the purpose of each .

Ans 4.

Tool	Purpose
Text Editor / IDE (VS Code)	Used to write and edit code.
Web Browser (Chrome, Firefox)	Used to run and test web pages.
Git	Tracks code changes and manages versions.
GitHub	Stores and shares code online.
Node.js	Runs JavaScript outside the browser and manages packages.
Package Manager (npm)	Installs and manages libraries and dependencies.
Local Server (XAMPP/Live Server)	Runs the website on the local computer for testing.
Browser Developer Tools	Debugs HTML, CSS, and JavaScript.

Ques 5 . Explain what a web server is and give examples of commonly used servers.

Ans 5 . A **web server** is software or hardware that stores website files and delivers web pages to users when they request them through a web browser using HTTP/HTTPS.

Commonly Used Web Servers

- Apache HTTP Server
- Nginx
- Microsoft IIS (Internet Information Services)
- LiteSpeed
- Apache Tomcat

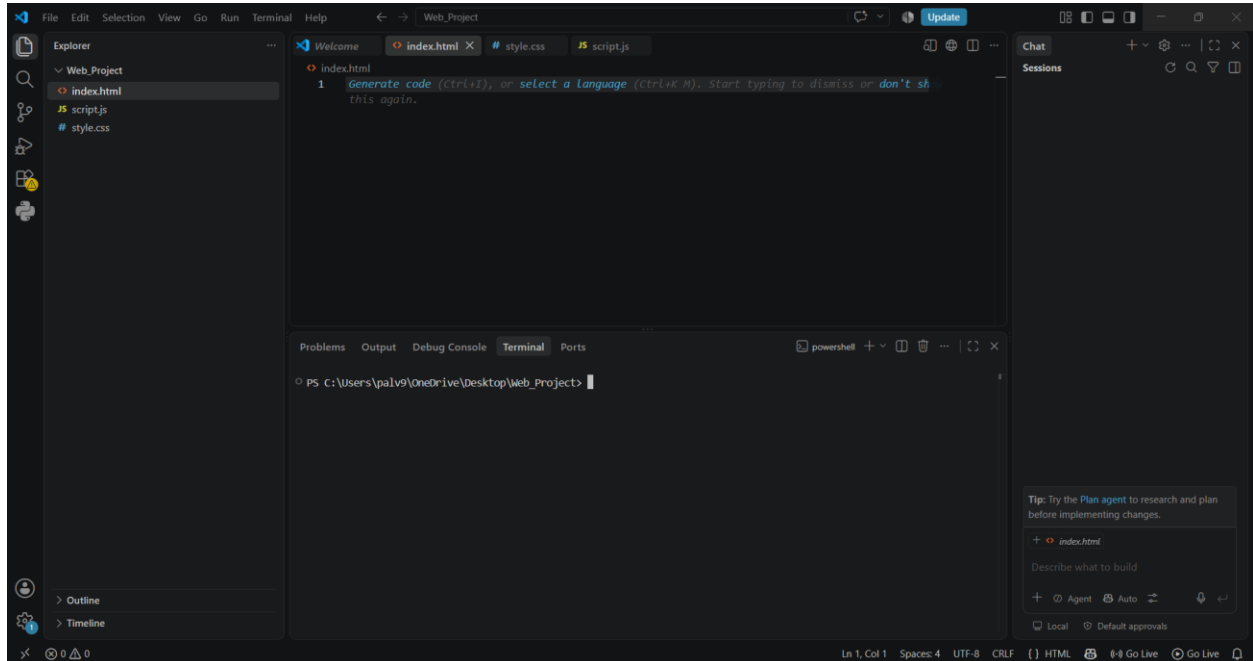
Ques 6 . Define the roles of a frontend developer, backend developer, and database administrator in a project.

Ans 6.

Role	Work
Frontend Developer	Designs and develops the user interface (UI) using HTML, CSS, and JavaScript .
Backend Developer	Develops the server-side logic, APIs, and handles communication between the frontend and the database.
Database Administrator (DBA)	Creates, manages, secures, and maintains the database to store and retrieve data efficiently.

Ques 7 . Install VS Code and configure it for HTML, CSS, and JavaScript development. Take a screenshot of the setup.

Ans 7 .



Ques 8. Explain the difference between static and dynamic websites. Provide an example of each.

Ans 8.

Static Website

Fixed content.

Does not use a database.

Simple and fast.

Example: Personal Portfolio, Company Website

Dynamic Website

Content changes based on user or database.

Uses a database.

Interactive and feature-rich.

Example: Amazon, Facebook, Instagram

Ques 9. ` Research and list five web browsers. Explain how rendering engines differ between them.

Ans 9.

Web Browser	Rendering Engine
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Google Chrome	Blink
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Mozilla Firefox	Gecko
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Microsoft Edge	Blink
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Apple Safari	WebKit
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Opera	Blink
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Rendering Engine :

A **rendering engine** is the part of a web browser that **reads HTML, CSS, and JavaScript and displays the webpage** on the screen.

Difference Between Rendering Engines

- **Blink** – Fast and widely used in Chrome, Edge, and Opera.
- **Gecko** – Used only by Firefox; focuses on web standards and privacy.
- **WebKit** – Used by Safari; optimized for Apple devices (macOS and iPhone).

Ques 10 . Draw a labeled diagram showing the basic web architecture flow — client, server, database, and APIs.

Ans 10 .

